Senior Design Final Report:
Musculoskeletal Assessment Device

EE 482: Electrical and Computer Engineering Design II

MAD

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ABSTRACT

This project was designed and developed in a way that allows people to accurately assess their muscle strength with and without the help of a healthcare provider. The importance of this project is that it provides people who need muscle rehabilitation with an easier way to track progress. This is accomplished via an electromyography (EMG) sensor. An EMG sensor senses the electrical signal sent to the muscles from the brain and quantifies it. The user will attach the EMG sensor and contract their bicep while holding a resistance band. If the person has greater muscle strength the brain will have to “recruit” less muscle fibers and will thus result in a smaller signal on the EMG. However, our final muscle rating score will use higher percentages to designate stronger muscles so that it is more intuitive for the user.

The result of this project proves that the design team has successfully created a prototype for a mobile device that can be deployed in the field or in patient’s homes that can measure muscle and joint strength for use in tracking growth, aging, and disease progression.

1. Background/Introduction

The need for any type of physical therapy assessment device is crucial for determining the state and progress of weak or injured muscles. The design team will be designing a Musculoskeletal Assessment Device (MAD) that accomplishes this. There are a handful of ways to go about designing this device. One method is determining muscle unit recruitment through the use of an electromyography (EMG) sensor. Muscle unit recruitment is how your body varies strength when using your muscles [1]. This is accomplished by changing how many muscle fibers are used or “recruited” for any given task. This project aims to use that fact in order to measure a person's muscular strength via an EMG sensor. An EMG or electromyography sensor measures the electrical activity produced by muscles [2]. EMG signals can be measured on the surface of the skin, under the skin, or even in muscle fibers. In order to keep this device as non-intrusive as possible, the skin surface method will be used. Using these two ideas, the design team will develop a portable and non-invasive approach to determining muscular strength in an individual for the purpose of tracking growth, aging, and disease progression.

For our design, the MSP-EXP430FR4133 (MSP) and sensors/other boards (excluding wiring) shown in Figures 1 and 2 will be used. The LED shield board will attach to the top of the muscle sensor, and the electrodes (connected to the sensor) will be temporarily attached to the patient through the use of sticky disposable pads as shown in Figure 2.

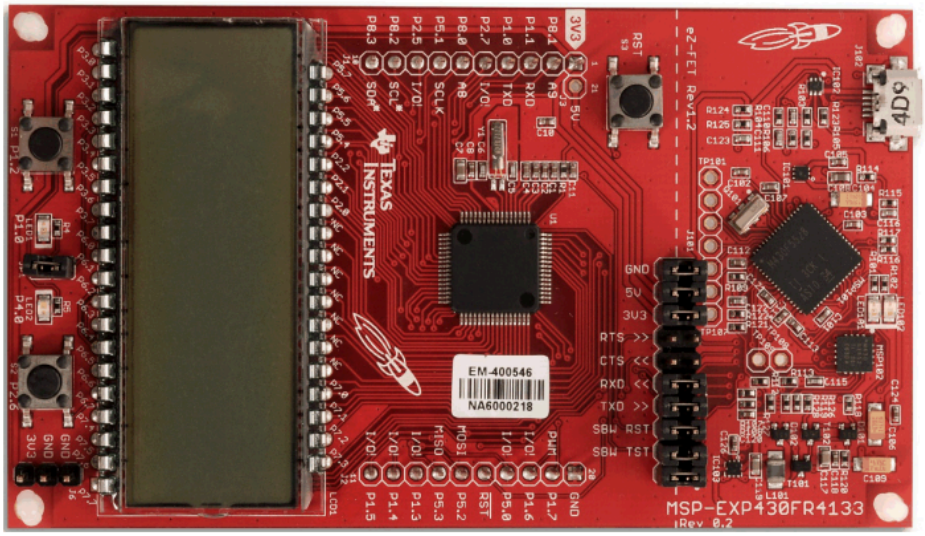


Figure 1: MSP-EXP430FR4133 Microcontroller used for MAD [3]

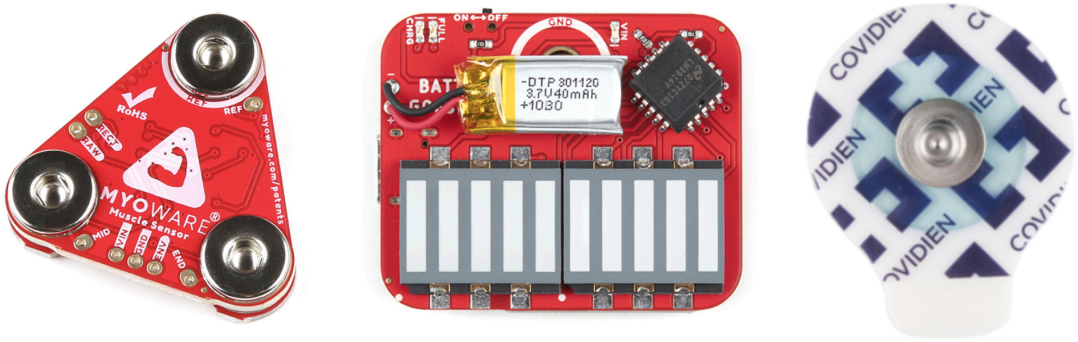


Figure 2: The MyoWare 2.0 Muscle Sensor (left), MyoWare 2.0 LED Shield board (middle), and 24mm Disposable Surface EMG/ECG/EKG Electrode Pads (right) [4].

This project has been designed in the most efficient and logical manner possible. The block diagram of the MAD is shown in Figure 3 below. The EMG sensor and electrodes need to be directly attached to the patient's skin attached to the patient via their skin contact, while being commanded by the microcontroller when to take measurements. The LED sensor is also attached directly to the EMG sensor, since its output is controlled directly by the voltage it senses through the electrodes. It also has its own battery pack, so it doesn't need to be attached to the power supply. Lastly, the LCD screen is built into the microprocessor, and both need to be powered by a 5 V power supply, which was accomplished with a rechargeable battery pack. The functional decomposition model and flowchart are shown in Figures 4 and 5 respectively.

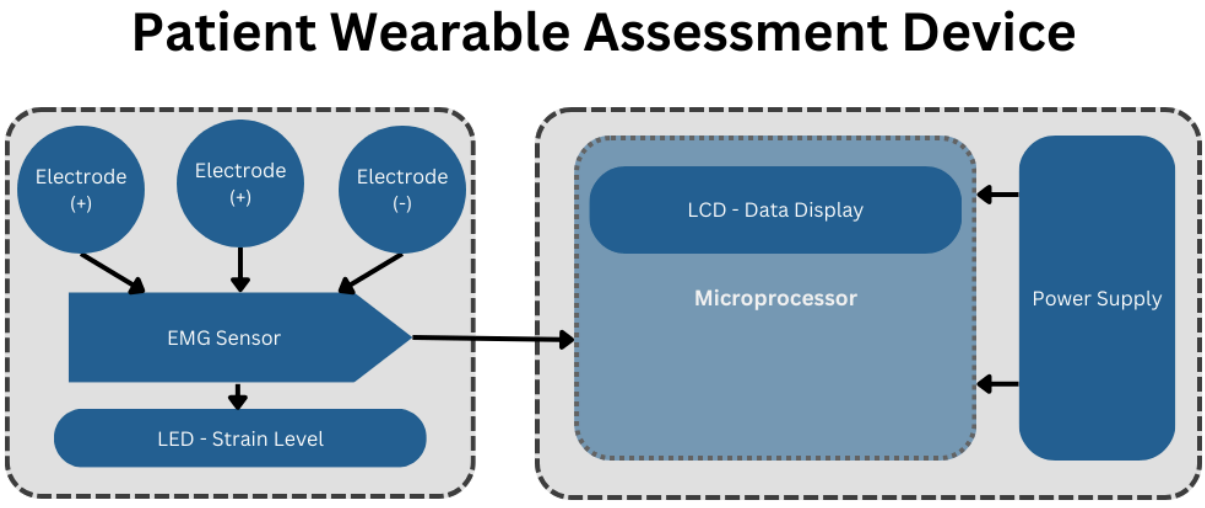


Figure 3: Block Diagram of MAD.

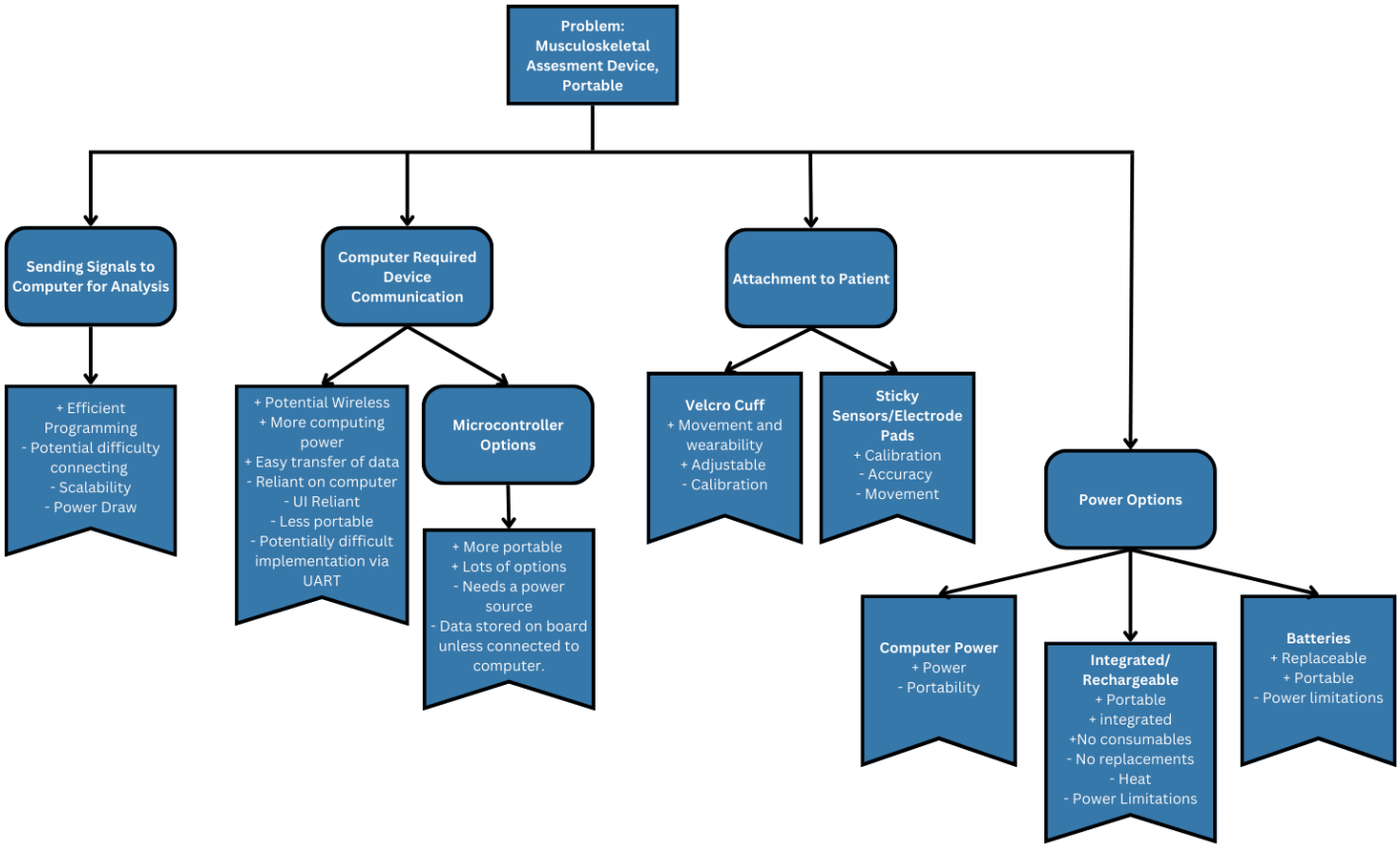


Figure 4: Functional Decomposition Model for MAD.

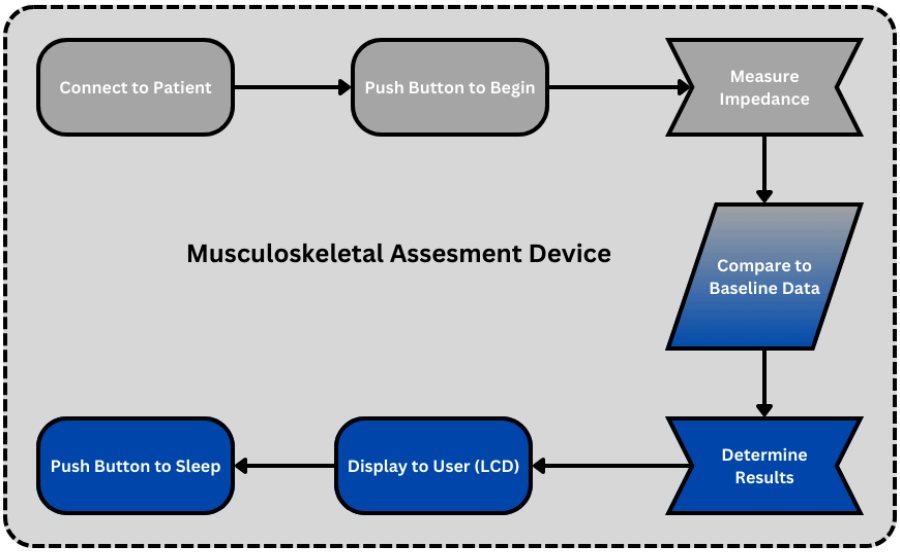


Figure 5: Flowchart Model for MAD.

There were a few sensor methods available for the MAD device that were decided against during the design process. These sensor methods include force myography (FMG), mechanomyography (MMG), and strain-mediated contact in anisotropically resistive structures (SCARS). SCARS sensors rely on changes in ohmic contact between adjacent restrained carbon fiber meanders to determine muscle deformation [5]. As the sensor strains, the meanders in the device separate, which increase overall resistance. Mechanomyography is the use of mechanical signals observable from the surface of a muscle when the muscle is contracted. It has a higher signal-to-noise ratio than the surface EMG and can be used to monitor muscle activity from deeper muscles without being too invasive [6]. Force myography is a lesser used option and is capable of using force sensors to register the variation of muscle stiffness patterns around a limb during different movements [7]. Overall, the EMG signal was chosen because it is non-invasive, reliable, and easier to process the data once collected. This scale is shown in Section 2.2, Table 2.

2. Design

The design of the MAD device primarily consists of configuring the analog-to-digital converter and LCD screen, along with engineering the physical cuff in a way that provides users ease of use each time a measurement is taken. Throughout the design of this device, the design team has deemed these requirements to be necessary:

- Patient wearable and non-invasive
- Portable (battery powered)
- Easy user interface
- Provide useful feedback
- Safe for the patient
- Reusable (disposable electrode pads)

2.1 Design Procedure

When it came down to the design procedure, realistic constraints and engineering tradeoffs had to be kept in mind so that a successful prototype could result.

2.1.1 Multiple Realistic Constraints and Engineering Tradeoffs

When designing the MAD, the team faced three major engineering constraints and each came with tradeoffs as listed in Table 1. The first constraint was power supply. The team needed some sort of way to power the device. The second constraint was the need for a device which could convert muscle measurements into something that a person can understand. The final constraint was which method of muscle assessment the team wanted to use.

In terms of power the team had two main options: micro USB or a battery pack. The team decided to use a battery pack because portability is a vital factor in our final product. The battery pack directly connected to the microcontroller and supplied voltage to our entire device.

When choosing a device that could convert the data from the sensor into numbers the end user can understand, the team decided to go with the MSP-EXP430FR4133. The team chose this microcontroller because it provided a good way to display the data collected via an LCD screen and it was relatively inexpensive given its price tag of $14. The tradeoff here is that given the cost of the device, the LCD screen is very limited in its display options which limits the type and amount of data that can be displayed. For reference, it closely resembles a seven segment display with 6 digits.

Finally, for the muscle assessment method, the team decided on using an Electromyography (EMG) sensor. The main factor for deciding on this method was that the EMG sensor was readily available for purchase. The tradeoff here is that what was gained in convenience was lost in accuracy since the EMG sensor only senses electrical signals on the surface of the skin. It is not as accurate as other methods such as the strain-mediated contact in the anisotropically resistive structures method or bioelectrical impedance analysis.

Table 1: Engineering Constraints and Tradeoffs for MAD.

	Constraints	Tradeoffs
Power	USB Powered	Less expensive but requires an external power source.
	Battery Powered	Much more portable, but added cost of batteries and battery casing.
Device	MSP430	Relatively cheap,overboard for application, higher power requirements, more familiar, has an LCD screen to display data.
	Arduino	More expensive, harder to implement modules like LCD screens and ADCs.
Method	EMG	Much easier to implement but less accurate.
	Bioelectrical Impedance Analysis	Harder to implement but more accurate.

2.1.2 Circuits and Systems Design Description

The circuit layout for the device was simplistic in that the single sensor was easily applied to the MSP430. This layout is shown in Figure 6. The EMG sensor only required three wiring sources which included the supplied power (red), a ground reference (green) and an analog data output (black) to the MSP. Recommendations for the circuit configuration were referenced from the EMG Sensor Guides [4].

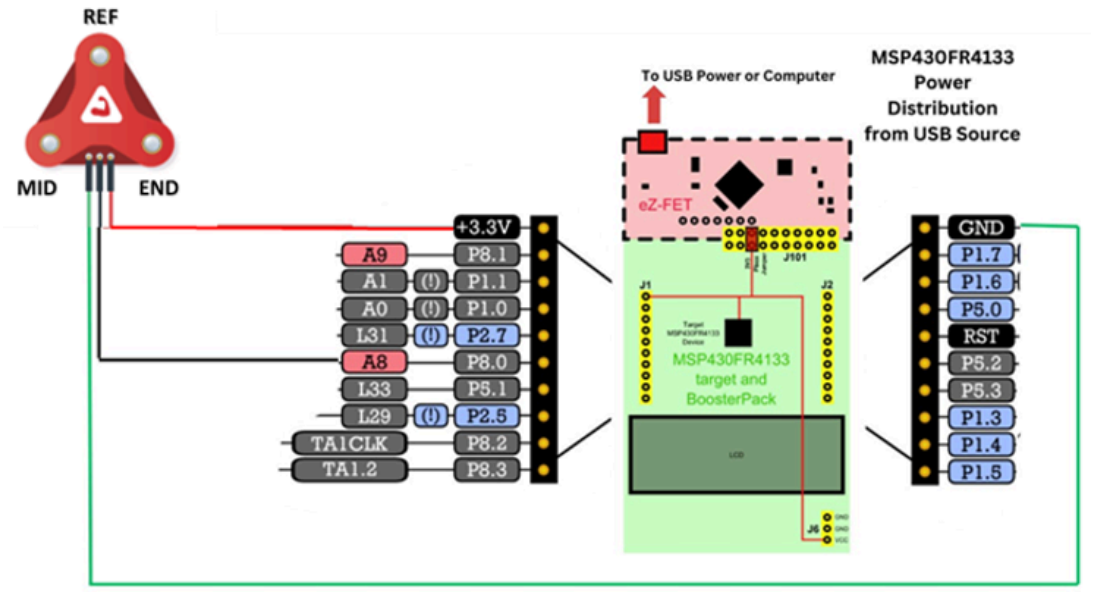


Figure 6: Schematic Diagram Without Power Supply.

The primary coding portion for this device requires implementation of an analog to digital converter (ADC), Figure 7 below contains the block diagram of the system's ADC. The ADC is internal to the MSP430 and capable of up to twelve individually configurable external input channels. For the purposes of this project, only one external input channel will be used for reception of the analog voltage signal associated with the EMG sensor output. Verification for the functionality of the ADC can be observed with the use of an oscilloscope capturing the voltage signal sent to the MSP430 from the sensor.

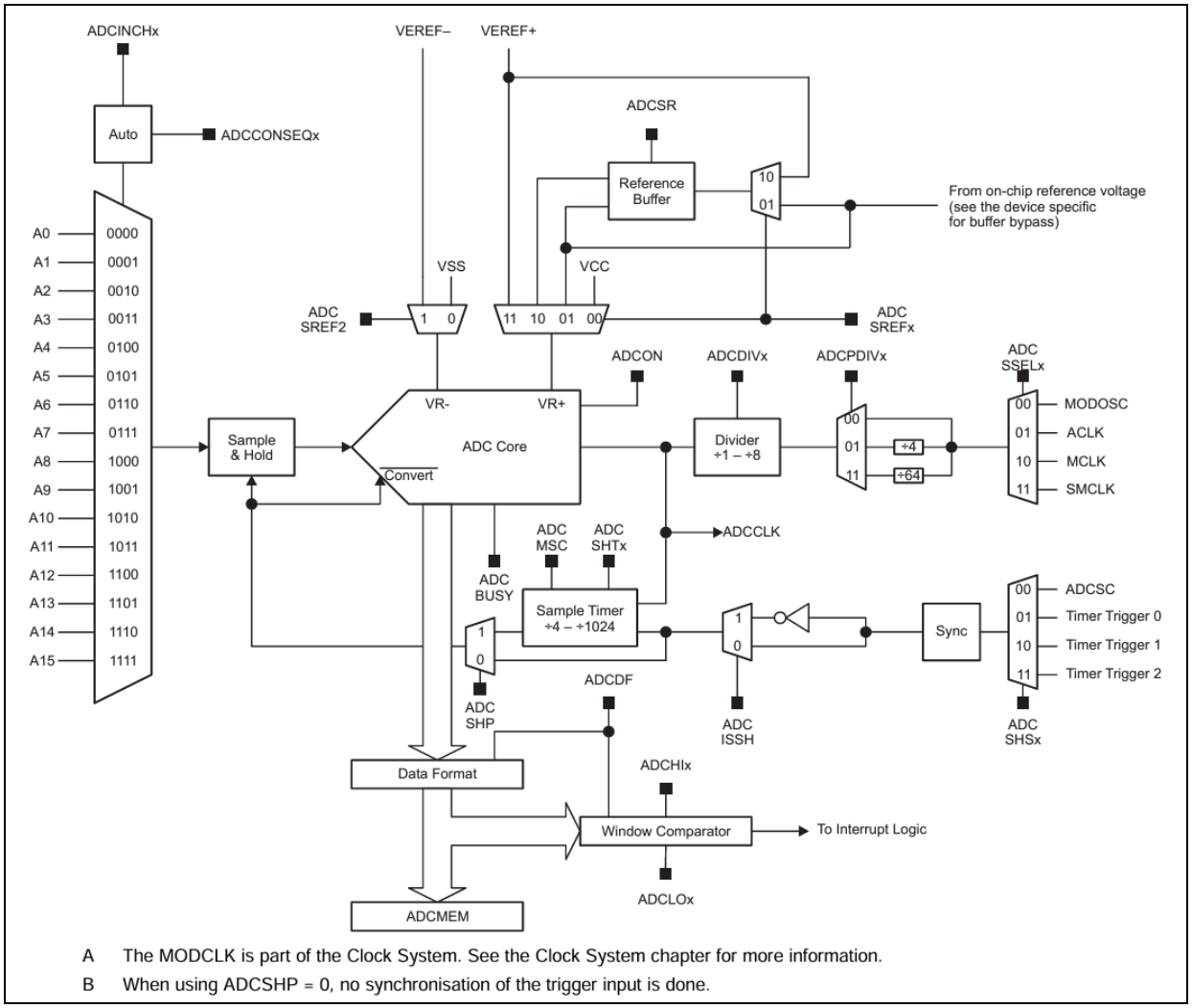


Figure 7: MSP430FR41xx User Guide ADC Block Diagram [3].

Figure 8 below shows a test signal caused by pressing the button on the MSP430. This signal triggers the ADC to start taking samples for conversion and for the array created within the program code to begin storing data. These values can then be read for hand calculations of sample signal voltages for comparison to the maximum signal voltage, which can confirm the sensor is sending the correct voltages to the MSP430. Sample code files from Texas Instruments helped to quickly understand the programming required for our project and delved into some of the more complicated modules of our code [8]. A more thorough analysis of the programming is explained below.

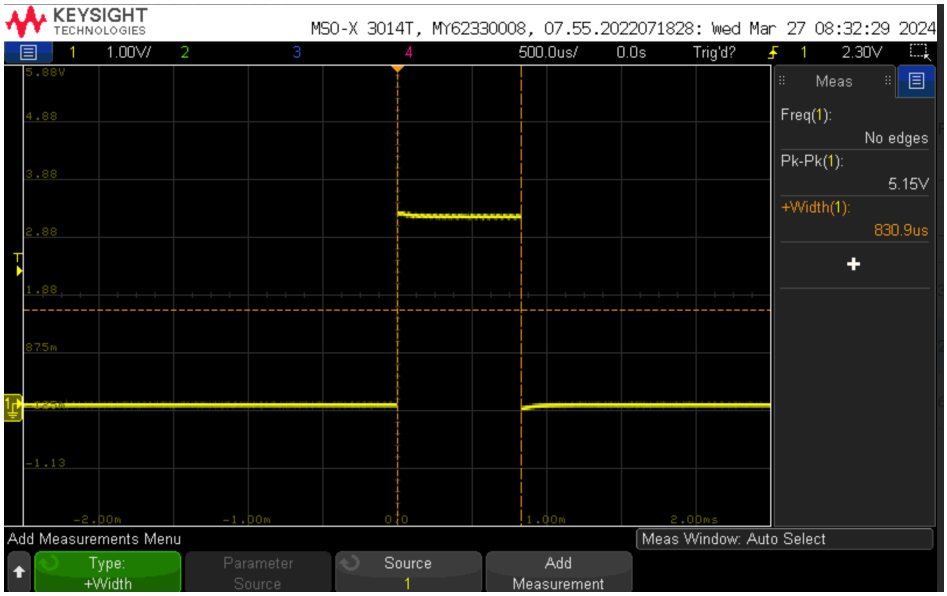


Figure 8: Test Trigger Signal from the Push Button Captured on the Oscilloscope.

Programming the ADC was facilitated by referencing the example files from Texas Instruments [8]. This MSP430 required the implementation of an array in the code to store samples taken from the EMG sensor as the ADC module did not have additional storage registers. The program code for the array specifications and ADC parameters can be seen in Figure 9 below.

```
// Configure ADC - Pulse sample mode; ADCSC trigger
ADCCTL0 |= ADCSHT_8 | ADCON + ADCMSC; // 256 clock cycles, ADC ON, multi sample/convert
ADCCTL1 |= ADCSHP + ADCCONSEQ_2; // s/w trig, single ch/conv, MODOSC
ADCCTL2 |= ADCRES; // 10-bit conversion results
ADCMCTL0 |= ADCSREF_0 | ADCINCH_8; // ADC input ch A8
ADCIE |=ADCIE0; // Enable the Interrupt request for a completed ADC_B conversion
```

Figure 9: ADC Parameters.

In Figure 9, it can be seen that the setting ADCSHT_8 was initialized. This setting controls the sample and conversion time for the ADC, specifically that it will take 256 clock cycles [13]. The team decided to leave the default clock sources as they were in the program because the reaction time of the human body is greater than the total time the program will collect samples. Equation 1 shows the time per sample that the ADC takes to collect and Equation 2 shows the total time to collect all 400 samples to fill the array elements.

Per sample:

$$\left(\frac{1}{4.8\ MHz}\right) \times 256\ clock\ cycles\ =\ 53.33\ us \qquad \text{Eq. 1}$$

All samples:

$$53.33\ us \times 400\ samples\ =\ 21.33\ ms \qquad \text{Eq. 2}$$

Figure 10 shows the initialization interrupt for the ADC, which begins when the button is pressed on the MSP430. When the button is pressed, the ADC begins reading analog inputs from the EMG Sensor.

```
void Button(void) __interrupt[PORT1_VECTOR]{ //start sampling when button pressed
//Check if button press happened
   if (P1IV == P1IV_P1IFG2) //Button press interrupt present for P1.2
   {
    if(buttonFlag == 0){
      ADCCTL0 |= ADCSC + ADCENC; //start conversion
```

Figure 10: Button Press Interrupt, ADC Starts Sampling, and Conversion.

Figure 11 shows what is done with the analog samples after the collection process begins. After the ADC begins receiving data, each data sample is passed into an array. The array is looped to allow for overflow data, however it is unlikely that the array will be completely filled. This is because the chosen number of elements, 400, is greater than the programmed sampling clock cycles of the ADC.

```
// ADC interrupt service routine
void ADC_ISR(void) __interrupt[ADC_VECTOR]{
  if(i < 400){
  ADC_Result[i] = ADCMEM0; //fill array of 400 elements with samples
  i++;
  __delay_cycles(10000); //delay to allow program to catch up
  }
  else{//input code here for when we're done taking our samples
  ADCCTL0 &= ~ ADCENC;
  mVal = 0;
    for(i = 0; i < 400; i++)
    {
        mVal += ADC_Result[i]; //sum all samples
    }
  mVal = mVal/400; //average samples
  i=0;
  LPM0_EXIT; //exit low power mode
  }
  P1IV = 0; //clear interrupt flag
}
```

Figure 11: Program Code to Fill Array with Input Samples.

Once all of the samples are collected into the array, the array elements are summed and averaged. This provides a value to be passed back into the main program for conversion into a percentage that is shown on the LCD.

2.2 Design Details

Initially, the team planned on using a voltage divider and battery back of four triple A batteries for the power supply. When it was tested, the LCD screen went dim and started flashing. The design team came to the conclusion that the batteries were not supplying enough current to the microcontroller. In our final design, the team decided on using a battery pack made for remote charging a smartphone with a micro USB connector. The battery pack was tested and deemed successful. The MSP430 Experimenter Board typically receives 5 V from the provided micro USB cable, when connected to a computer, this is then regulated within the board down to 3.3 V for the microprocessor and also distributed throughout the other board components. A single triple A battery is capable of producing 1.5 volts and 50 milliamps for 1200 milliamp-hours. The four batteries arranged in a series could produce up to 6 volts and 50 milliamps for 1200 milliamp-hours. Equation 3 shows the voltage calculation for the initial designed battery pack and Equation 4 shows the current calculation.

$$Series\ Voltage\text{: } V_1 + V_2 + V_3 + V_4 = V_T \Rightarrow 4(1.5\ V) = 6\ V \qquad \text{Eq. 3}$$

$$Series\ Current\text{: } I_1 = I_2 = I_3 = I_4 = I_T \Rightarrow 50\ mA \qquad \text{Eq. 4}$$

The commercially available rechargeable battery pack provides 5 V for 2900 milliamp-hours. The current provided by this pack is 2.4 A, this is well within the range required to operate the MSP430 Experimenter Board and any peripheral devices attached.

The MYOWARE 2.0 LED Shield came with its own battery that supplies the board with 3.7 V. The MSP430 provides the sensor with 3.3 V, so the team was worried about the LED shield battery overpowering the microcontroller and/or sensor. The design team decided to disconnect the power terminal from the LED shield to the sensor with electrical tape as a result.

In our final program shown in Appendix D, the design team chose to display “PRESS BUTTON” across the LCD screen when the microcontroller is turned on in order to direct the user to begin the measurement once ready. The two memory registers for the LCD screen are used for this action (since “PRESS” and “BUTTON” are flashed back and forth) and are cleared after the user presses the button. After this, the word “MEAS” (short for measuring) is displayed across the LCD screen during the measurement process. The words move from the left of the screen to the right, signaling that the device is working. The program then takes an average of the voltage recorded from the sensor over a time span of about 5 seconds and displays the number out of 100 to the LCD screen. The team decided to have higher scores represent less muscle unit recruitment, meaning that higher percentages are resultant of a stronger muscle. Table 2 shows the metric we developed to categorize measurement. These values were based on the maximum possible sample resolution of the ADC. Sample resolution of the ADC is maximized at the supply voltage of approximately 3.3 volts, which is analogous to the ADC value of 1024.

Table 2: Developed Measurement Metric Based on ADC Resolution.

% shown	ADC value	Analysis
0%	1013.76	Possible Muscle Failure, Seek Medical Attention
10%	921.60	Poor Muscle Engagement (Over Exertion)
20%	819.20	
30%	716.80	Average Muscle Engagement
40%	614.40	
50%	512.00	
60%	409.60	
70%	307.20	Excellent Muscle Engagement
80%	204.80	
90%	102.40	
99%	0.00	

Mounting options for the MSP430 and its power supply were developed with assistance from UAF's Makerspace and the use of their 3D printers. A stabilization mount was incorporated to add rigidity to the MSP430 and create a place for locating the power supply close to the microcontroller. A 'pass-though' was incorporated into the cuff for feeding wires from the MSP430 to the EMG sensor via velcro that is located on the inside of the cuff, closest to the patient's skin. In addition, an easy access velcro panel was designed to protect the MSP and allow accessibility to the batteries within the power supply. This physical cuff is shown in Figure 12.

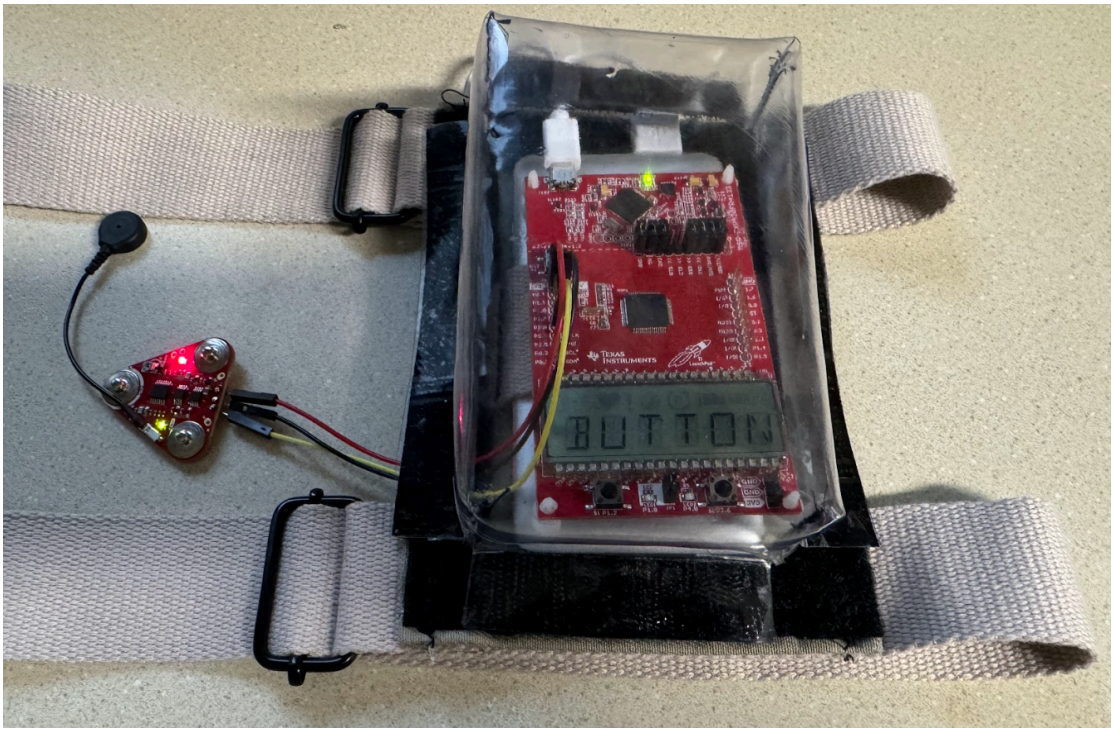


Figure 12: Physical MAD.

2.3 Application of Engineering and Medical Standards

Engineering standards are important in maintaining and ensuring the safety and reliability of electrical devices. For the MAD, there are pertinent IEEE standards to the quality and reliability of our design, which include 1063-2001, 1012-2016, and 1073.3.2-2000 as well as IEC 80002-1:2009, and IEC 60601-1-2:2014.

2.3.1 IEEE 1063-2001

IEEE 1063-2001 states: “The organization of documentation shall support its usage mode … and shall provide complete instructional and reference information for all critical software functions…including all instances of the selected elements being documented and covers a subset of software commands…[9]”. Our documentation has been meticulously designed to meet the necessary standards for reference documentation, ensuring easy and quick access to information whenever needed. Given the extensive focus on documenting the creation and usage of the device, the documentation caters to the requirement of accessing information conveniently and without any specific order. By adhering to the organizational standards of IEEE 1063-2001, the design team can provide a reliable guide for the use of our device.

2.3.2 IEEE 1012-2016

IEEE 1012-2016 states: “Successfully demonstrates whether the requirements are correct, complete, accurate, consistent, and testable…evaluating source code components for validity that software code and its interactions with other elements do not result in unnecessary, unintended, or deleterious consequences…verify that all terms and code concepts are documented consistently [10].” Throughout all phases of construction, the team will rigorously assess the requirements and how they are obtained to determine whether the requirements are correct, complete, accurate, consistent, and testable, as stipulated by the standard. Additionally, the team will diligently evaluate source code components to verify that our software code and its interactions with other elements are complete, and documented through to terms and code concepts to ensure the user’s safety and clarity for the reader. By adhering to IEEE 1012-2016, our device will undergo several weeks of scrutiny, resulting in accurate and complete specifications and ultimately providing a reliable product.

2.3.3 IEEE 1073.3.2-2000

IEEE 1073.3.2-200 states: “...the maximum cable length of 20m..it is suggested that the powered device or adapter a) operate normally if voltage and current are sufficient to function correctly, b) shut down if power is insufficient to function correctly c) simply not function at all if there is little or no power [11].” The team’s design and implementation adhere closely to these specifications. The limitation on cable length ensures optimal performance within the specified parameters, while the suggested behavior of the powered device reflects considerations for safety and reliability, which can be implemented in both hardware and software adjustments. By implementing these recommendations, the team can ensure that the device operates reliably under various conditions, maximizing both performance and safety.

2.3.4 IEC 80002-1:2009

IEC 80002-1:2009 states: “...complex software designs can permit complex sequences of events which may contribute to hazardous situations….identifying points in the sequences of events at which the sequence can be interrupted, preventing harm or reducing its probability…unforeseen software responses to input…and events arising from incorrect coding…should be evaluated on the severity of the harm alone [14]. Throughout the development of the program utilized by the MAD, the potential of harmful software setup has been considered and evaluated with careful consideration to the potential outcome if input is not as expected or some portion of the code is incorrectly implemented. By keeping IEC 80002 in mind, the team can ensure that the code is simplified, and written as to prevent potential risk to a patient due to the manner in which inputs are to be handled.

2.3.5 IEC 60601-1–2:2014

IEC 60601-1-2:2014 states: "...medical electrical equipment and medical electrical systems are expected to provide their basic safety and essential performance without interfering with other equipment and systems in the electromagnetic environment in which they are intended… [15]." Through careful design and planning, the team has considered the potential for outside interference of the MAD and has accommodated this by considering the intended environment of use, which has little to no other electrical systems present.

In adhering to IEEE standards such as 1063-2001, 1012-2016, and 1073.3.2-2000, as well as IEC 80002-1:2009, and IEC 60601-1-2:2014, the team is committed to producing a MAD that prioritizes safety, reliability, and user-friendliness. Through meticulous documentation, rigorous evaluation of requirements and source code, and careful consideration of device behavior under various conditions, the team aims to deliver a product that not only meets but exceeds industry standards, ensuring the utmost satisfaction and confidence for the users.

3. Verification and Testing

When it came to verification and testing there were two parts the team tackled: the components and the programming that would connect them.

3.1 Testing and Verification of Components

When our materials arrived, the design team tested the sensor with the accompanying plug and play module, MYOWARE 2.0 LED Shield. It was verified that the sensor works in accordance with the factory provided datasheet [4]. When the subject flexed against the resistance band, the sensor indicated an associated voltage shown with the LED array on the LED Shield, the testing example is shown in Figure 13 below.

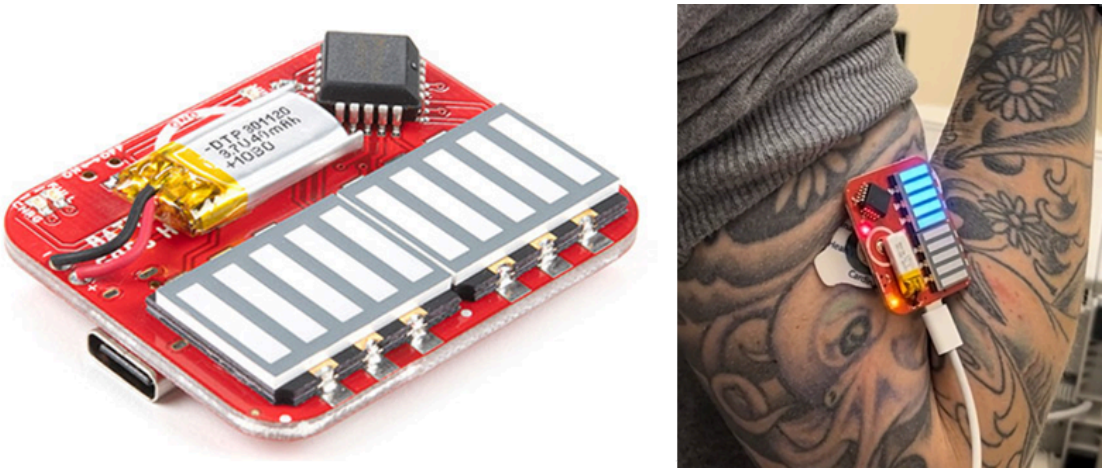


Figure 13: MYOWARE 2.0 LED Shield for MYOWARE 2.0 EMG Sensor Testing [4].

3.2 Testing and Verification of Program

The microcontroller was shipped with outdated firmware from Texas Instruments, this caused the microcontroller to not respond to the required software for programming. The firmware issue was remedied with the installation of updated firmware and the use of proprietary software from Texas Instruments on Duckering 210 lab computers and help from Professor Dejan Raskovic. This solution was preferred, as administrator privileges are required to install software on university computers. Our finalized current requirements and verifications for the entire MAD device is shown in Table 2.

Table 3: Requirements and Verifications for Current Device Setup for MAD.

T = Testing and Measurements A = Analysis and Simulation O = Observation and Inspection R = Reference and Datasheet

Number	Requirement	Verification Detail	T	A	O	R	Comments
1.0	Microcontroller	The microcontroller will oversee device functionality and operations.	X	X	X	X	Established comms with PC, begin writing program.
1.1	LCD Screen	The LCD Screen will display instructions for the user and device measurements when applicable.		X	X	X	N/A
1.2	Communication	The microcontroller will communicate with the muscle sensor quickly and accurately.	X	X	X	X	N/A
2.0	Muscle Sensor	The muscle sensor will produce relevant and accurate data from the user's bicep.	X	X	X	X	Preliminary out of box testing completed, sensor is functional.
2.1	LED Board	Displays the measurements from the muscle sensor in a readable manner for the user.	X	X	X		Displays a range of LEDs based on the effort required.
2.2	Waveform Output	The device will accurately sense muscle activity and produce an accompanying waveform voltage reference (on MATLAB if the user desires).					MATLAB was not incorporated due to time constraints.
2.3	Weight	The sensor will weigh less than 50 grams so that it can stick comfortably to the patient's arm without risk of disconnection.	X		X		Total system weight verifications upon completion of the initial construction.
2.4	Battery	The device will be mobile and battery-powered to be deployed in patient homes.		X	X	X	Provides MAD with voltage needed to operate.

4. Estimated Cost

With all components ordered, and the prototype MAD built, the final estimated costs can be evaluated.

4.1 Total Budget

With the application for funding towards the MAD, the team was awarded a total of $504.26 from Center ICE. With this funding in mind, materials were able to be purchased to begin the building process.

4.2 Cost Breakdown

The project costs are divided into two sections: the cost of the components and resulting labor costs.

4.2.1 Purchased Component Cost and Shipping

Based upon the component cost breakdown listed in Table 4, components were ordered and received. The list of components has remained primarily unchanged as the sensors, cables, pads, connectors, and MSP microcontroller board were all obtainable, although some components had a longer delivery timeline. In addition to the baseline components that are listed in Table 4, several items were not considered in the initial budget including batteries to power the device and battery holders which were not built onto the MSP board, which can be seen at the bottom of the Table 4.

Table 4: Component List for MAD with Part Numbers and Cost.

COMPONENT	PART NUMBER	COST ($)
MyoWare 2.0 Muscle Sensor. all-in-one electromyography (EMG) sensor	DEV-21265	$39.95
MSP-EXP430FR4133 Breakboard for sensor main connection	MSP-EXP430FR4133	$16.79
Disposable Surface EMG/ECG/EKG Electrode - 24mm (10 pack)	SEN-12969	$8.95
MyoWare 2.0 Reference Cable (10 pack)	KIT-19627	$10.95
Sensor Cable - Electrode Pads (3 connector) for MyoWare Muscle Sensor	CAB-12970	$5.50
MyoWare 2.0 LED Shield	DEV-18387	$26.95
AA Batteries(10 pack)	-	$12.79
3 AA Battery Holder(4 pack)	-	$7.99
Total Cost		**$129.87**

Before factoring in shipping costs, the total list of components in Table 4 resulted in a cost of $129.87, not including cost adjustments. In all orders that had to be performed to acquire the components needed for the device, Tariff as well as shipping and handling charges were added onto the price of the components. The resulting additional charges for each set of component orders differed by vendor, and resulted in a total of $32.09 as displayed in Table 5.

In addition to the total additional charges that were placed on all orders of parts, the disposable surface electrodes were ordered in excess, as these components had multiple potential uses, and were limited to a single person's use. Due to this, the total cost of components alone displayed in Table 5 resulted in $172.44, which was 34.2% of our total funding allotment. When considering the total value of components along with the additional charges, the total came to $204.5,3 which was 40.5% of our total funds.

Table 5: Ordered Items List and Corresponding Charges for MAD.

Vendor	Ordered Items (quantity)	Total Component Costs	Additional Charges
Digikey	MSP430FR4133 Board	$16.79	$8.67 (Tariff/shipping)
Mouser Electronics	MyoWare 2.0 Muscle Sensor (2)	$79.90	$7.99 (freight)
SparkFun	Sensor Cables - Electrode Pads (2) MyoWare 2.0 LED Shield (1) MyoWare 2.0 Reference Cables - 10 pack (1) Disposable surface Electrode- 10 pack (3)	$75.75	$15.43 (shipping/handling)
Total Cost		**$172.44**	**$32.09**

4.2.2 Labor Cost (Actual Hours)

With an assumed rate of $35 an hour, the approximate hours came to be 40 hours of labor for each design team member, resulting in a total labor cost of $1400 respectively for each design team member as listed in Table 6.

Table 6: Labor Hours and Cost Breakdown for MAD.

Person	Rate/Hour	Current Hours	Duration Total
Sheridan Blasey	$35/ hr	40 hr	$1400
Lisa Jacklin	$35/ hr	40 hr	$1400
Elizabeth Lee	$35/ hr	40 hr	$1400
Daniel Kling	$35/ hr	40 hr	$1400
Total Labor Cost		**$ 5600**	

4.3 Current Total Cost

With the total labor cost as well as components and charges to obtain these components, the total estimated cost comes out to be $2304.53. A breakdown of the total cost by subsections can be found in Table 7 below.

Table 7: Total Cost Breakdown for MAD.

Cost Section	Total Value
Total Current Labor	$5600.00
Total Current Components	$172.44
Total Unexpected Charges	$ 32.09
Overall Total	**$5804.53**

5. Conclusions and Recommendations

Even though the team faced some challenges related to this project, the device was developed as described in our proposal in November 2023. When developing the MAD there were two main features that the team felt were most important: portability and ease of use. The portability aspect was important because it opens the door for this product to be used anywhere in the world whether that be in the field, the doctor's office or the comfort of one's own home. In addition, ease of use was important to the team because it allows anyone to use the MAD to see the progress they are making in their physical therapy or workout journey. Ultimately, the design team has successfully constructed a working prototype. Future adjustments to the prototype could be added if this project was continued, such as using a different muscular assessment method, SCARs for example, or adding a data logging capability so that users don’t have to keep track of past measurements.

5.1 Overall Design Results and Accomplishments

The main accomplishment for the team this semester was the completion of the senior design project. The team created a device that is both portable and takes valid measurements of a person's muscular strength which could be compared against Table 2 above. That said, there were a few notable challenges faced by the team when developing the MAD. The first was the difference between the MSP430F5438A and MSP430F4133. Originally when ordering our microcontroller, the team assumed that the onboard modules between both MSP430’s would

be the same, but there ended up being some key differences that the team had to overcome especially in regards to the ADC and LCD modules. For example, the MSP430 that the team purchased did not have more than one ADC memory register, which forced us to enter an interrupt every time the device took a sample. Whereas on the other MSP430 the device can take a sequence of samples before having to service an interrupt. That said, the difference in MSP430 issue was remedied partially thanks to some examples provided by Texas Instruments on their website and partially due to some laboring over the data sheet. Another issue the team faced was with the battery pack that was developed. The battery pack employed 4 AAA batteries in series to power the microcontroller over a micro USB cable. It turned out that the microcontroller and attached sensor needed more current than the batteries could provide thus we had to switch to a portable charger to meet the requirements.

5.2 Ethical Considerations

The IEEE Code of Ethics serves as a reference for ethical issues in this project. The complete code is fundamental to the design of the MAD. However, as a device with medical applications, the two most important clauses of the code for the project are as follows.

IEEE Code of Ethics Statement 1: “To hold paramount, the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;” [12]

IEEE Code of Ethics Statement 5: “To seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;” [12]

When developing MAD, the team prioritized safety and adhered to rigorous testing and quality assurance. Transparency in documentation is crucial, especially in the full functionality of MAD. These principles align with both Articles 1 (public safety, health and welfare) and Article 5 (prompt disclosure of factors endangering the public) of the IEEE Code of Ethics.

5.3 Broader Impacts

With use of an at-home physical therapy device brings the risk of lack of proper care to patients. If the device is marketed in a way that seems to supersede doctors and health care workers, patients could miss out on vital physical assessment. Upon completion of this project, it needs to be ensured that the purpose of this device is to assist medical professionals and not replace

them. Economically, this could mean that the device can be sold and targeted towards assisting the medical field as a part of physical therapy treatment. As long as the purpose of the device is clear, advertising to this demographic does not appear to have any negative economic effects. If this device becomes commercialized, it will be the design team's priority to ensure that the design team can market the product for as low as possible (while still making money). Since the device is an essential muscle rehabilitation device and could potentially change patient's lives, the design team wants everyone to have access to this device.

This project uses a portable, rechargeable battery for ease of use. The design team will make sure that the device utilizes low power modes as much as possible to increase the battery life so that it doesn't have to be recharged as frequently. . This project does incorporate one-time use disposable pads which could contribute to higher waste levels, especially if this project was incorporated into daily life across the world. If this was the case, the design team could look into reusable sticky electrode pads to minimize this. But, this may not be permissible because of sanitary reasons.

Keeping these impacts in mind, time has been spent making sure our project can have a positive impact upon completion. Since this project is focused around the rehabilitation and physical therapy treatment of a patient's muscles, the design team needs to ensure that the data recorded is collected and analyzed correctly. If the device has conveyed positive trends (muscle recovery) to the user, it must accurately reflect the state of the muscle so that the patient does not engage in activities that they are not capable of. This will keep the patients safe and healthy, and is a priority in the project.

5.4 Future Work

Future work on this project could go in a couple different directions. To start, the muscular assessment method could be changed to one that is more accurate such as strain-mediated contact in anisotropically resistive structure (SCARS). The team chose not to pursue this route when originally designing the device due to complexity and time constraints, but it would do a lot to improve the overall viability of the device. Given that this device is supposed to track the users muscular strength over a period of time, having some sort of data logging capabilities on the device to show the overall progress that the person has made would be beneficial. That said, the data does not necessarily need to be stored on board the MAD. If the data was transmitted to a computer wirelessly or via USB, it could be displayed in a format that is easier to understand such as a graph. This would make it easier to see the overall progress or trends in the user's data. Currently the team is in talks with Center ICE to do a "Invention Disclosure". This

will essentially hand off any future work done on this project to Center ICE, and if anyone decides that they wish to improve this project and monetize it, then the team will get a percentage of the royalties.

References

[1] F. Sandbrink, *“Motor Unit Recruitment in EMG: Definition of Motor Unit Recruitment, Order of Recruitment, Assessment of Recruitment at Low Level of Muscle Contraction”.* Sept. 2022. eMedicine, Available at*:* https://emedicine.medscape.com/article/1141359-overview#showall.

[2] Kamavuako EN. *On the Applications of EMG Sensors and Signals. Sensors (Basel).* 2022 Oct 19;22(20):7966. doi: 10.3390/s22207966. PMID: 36298317; PMCID: PMC9611382.

[3] *“MSP430FR4133 LaunchPad™ Development Kit (MSP-EXP430FR4133) User’s Guide MSP430FR4133 LaunchPad™ Development Kit (MSP--EXP430FR4133) MSP430FR4133 LaunchPad™ Development Kit (MSP-EXP430FR4133),”* 2014*.* Accessed: Apr. 15, 2024. [Online]. Available: https://www.ti.com/lit/ug/slau595b/slau595b.pdf?ts=1667054835064&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FMSP430FR4133

[4] *“MYOWARE® 2.0 Muscle Sensor,”* MYOWARE by Advancer Technologies. https://myoware.com/products/muscle-sensor/

[5] J. T. Alvarez et al*., "*Towards Soft Wearable Strain Sensors for Muscle Activity Monitoring,*" in IEEE Transactions on Neural Systems and Rehabilitation Engineering,* vol. 30, pp. 2198-2206, 2022, doi: 10.1109/TNSRE.2022.3196501.

[6] *“*Mechanomyogram*”, Wikipedia,* 23 March 2022 https://en.wikipedia.org/wiki/Mechanomyogram

[7] Z. G. Xiao and C. Menon, *“A Review of Force Myography Research and Development,” Sensors,* vol. 19, no. 20, p. 4557, Jan. 2019, doi: https://doi.org/10.3390/s19204557.

[8] “MSP430FR413x, MSP430FR203x code examples”, *Texas Instruments*, https://www.ti.com/tool/download/SLAC625 (accessed Apr. 14, 2024).

[9] *"*IEEE Standard for Software User Documentation*," in IEEE Std 1063-2001 ,* vol., no., pp.1-24, 20 Dec. 2001, doi: 10.1109/IEEESTD.2001.93368.

[10] "IEEE Standard for System, Software, and Hardware Verification and Validation," *in IEEE Std 1012-2016* (Revision of IEEE Std 1012-2012/ Incorporates IEEE Std 1012-2016/Cor1-2017) , vol., no., pp.1-260, 29 Sept. 2017, doi: 10.1109/IEEESTD.2017.8055462.

[11] "IEEE Standard for Medical Device Communications - Transport Profile - IrDA Based - Cable Connected," *in IEEE Std 1073.3.2-2000* , vol., no., pp.1-76, 14 April 2000, doi: 10.1109/IEEESTD.2000.91312.

[12] “IEEE code of Ethics,” *IEEE,* https://www.ieee.org/about/corporate/governance/p7-8.html (accessed Feb. 5, 2024).

[13] *“*MSP430FR4xx and MSP430FR2xx Family User’s Guide.*” Texas Instruments.* https://www.ti.com/lit/pdf/slau445. (accessed Jan. 24, 2024)

[14] “IEC 80002-1:2009 | IEC Webstore | Medical device software,” *webstore.iec.ch.* https://webstore.iec.ch/preview/info_iec80002-1%7Bed1.0%7Den.pdf (accessed Apr. 29, 2024).

[15] “IEC 60601-1-2:2014 | IEC Webstore | electromagnetic compatibility, EMC, smart city,” *webstore.iec.ch.* https://webstore.iec.ch/publication/2590

[16] Texas Instruments Incorporated, MSP430FR4133 LaunchPadTM Development Kit (MSP-EXP430FR4133) User’s Guide. 2017. https://www.ti.com/lit/ug/slau595b/slau595b.pdf?ts=1714418937291&ref_url=https%253A%252F%252Fwww.ti.com%252Ftool%252FMSP-EXP430FR4133. (accessed Jan. 24, 2024)

Appendix

Appendix A: MAD Final Program

```
#include <msp430.h>

unsigned int ADC_Result[512];
unsigned int i = 0;
unsigned int mVal = 0;
unsigned int buttonFlag = 0;

volatile unsigned char dataVal = 0;

void displayData(char dataVar);
void showChar(char c, int position);
void showCharB(char c, int position);
#define pos1 4                                  // Digit A1 - L4
#define pos2 6                                  // Digit A2 - L6
#define pos3 8                                  // Digit A3 - L8
#define pos4 10                                  // Digit A4 - L10
#define pos5 2                                  // Digit A5 - L2
#define pos6 18                                  // Digit A6 - L18
// Define word access definitions to LCD memories
#define LCDMEMW ((int*)LCDMEM)
#define LCDBMEMW ((int*)LCDBMEM)

// LCD memory map for numeric digits
const char digit[10][2] =
{
   {0xFC, 0x28},  /* "0" LCD segments a+b+c+d+e+f+k+q */
   {0x60, 0x20},  /* "1" */
   {0xDB, 0x00},  /* "2" */
   {0xF3, 0x00},  /* "3" */
   {0x67, 0x00},  /* "4" */
   {0xB7, 0x00},  /* "5" */
   {0xBF, 0x00},  /* "6" */
   {0xE4, 0x00},  /* "7" */
   {0xFF, 0x00},  /* "8" */
   {0xF7, 0x00}   /* "9" */
};

// LCD memory map for uppercase letters
const char alphabetBig[26][2] =
```

```
{
   {0xEF, 0x00},  /* "A" LCD segments a+b+c+e+f+g+m */
   {0xF1, 0x50},  /* "B" */
   {0x9C, 0x00},  /* "C" */
   {0xF0, 0x50},  /* "D" */
   {0x9F, 0x00},  /* "E" */
   {0x8F, 0x00},  /* "F" */
   {0xBD, 0x00},  /* "G" */
   {0x6F, 0x00},  /* "H" */
   {0x90, 0x50},  /* "I" */
   {0x78, 0x00},  /* "J" */
   {0x0E, 0x22},  /* "K" */
   {0x1C, 0x00},  /* "L" */
   {0x6C, 0xA0},  /* "M" */
   {0x6C, 0x82},  /* "N" */
   {0xFC, 0x00},  /* "O" */
   {0xCF, 0x00},  /* "P" */
   {0xFC, 0x02},  /* "Q" */
   {0xCF, 0x02},  /* "R" */
   {0xB7, 0x00},  /* "S" */
   {0x80, 0x50},  /* "T" */
   {0x7C, 0x00},  /* "U" */
   {0x0C, 0x28},  /* "V" */
   {0x6C, 0x0A},  /* "W" */
   {0x00, 0xAA},  /* "X" */
   {0x00, 0xB0},  /* "Y" */
   {0x90, 0x28}   /* "Z" */
};

void displayData(char dataVar){
LCDMEMW[pos3/2] = digit[dataVar/10][0] | (digit[dataVar/10][1] << 8);
LCDMEMW[pos4/2] = digit[dataVar%10][0] | (digit[dataVar%10][1] << 8);
LCDBMEMW[pos3/2] = digit[dataVar/10][0] | (digit[dataVar/10][1] << 8);
LCDBMEMW[pos4/2] = digit[dataVar%10][0] | (digit[dataVar%10][1] << 8);
}

//Writes to screen 1
 void showChar(char c, int position)
 {
    if (c == ' ')
    {
       // Display space
       LCDMEMW[position/2] = 0;
```

```
    }
    else if (c >= '0' && c <= '9')
    {
        // Display digit
        LCDMEMW[position/2] = digit[c-48][0] | (digit[c-48][1] << 8);
    }
    else if (c >= 'A' && c <= 'Z')
    {
        // Display alphabet
        LCDMEMW[position/2] = alphabetBig[c-65][0] | (alphabetBig[c-65][1] << 8);
    }
    else
    {
        // Turn all segments on if character is not a space, digit, or uppercase letter
        LCDMEMW[position/2] = 0xFFFF;
    }
}

//Writes to screen 2
void showCharB(char c, int position)
{
    if (c == ' ')
    {
        // Display space
        LCDBMEMW[position/2] = 0;
    }
    else if (c >= '0' && c <= '9')
    {
        // Display digit
        LCDBMEMW[position/2] = digit[c-48][0] | (digit[c-48][1] << 8);
    }
    else if (c >= 'A' && c <= 'Z')
    {
        // Display alphabet
        LCDBMEMW[position/2] = alphabetBig[c-65][0] | (alphabetBig[c-65][1] << 8);
    }
    else
    {
        // Turn all segments on if character is not a space, digit, or uppercase letter
        LCDBMEMW[position/2] = 0xFFFF;
    }
}
```

```
int main(void)
{
   WDTCTL = WDTPW | WDTHOLD;                          // Stop WDT

   // Configure GPIO
   P1DIR |= BIT0;                                  // Set P1.0 to output direction
   P1OUT &= ~BIT0;                                   // Clear P1.0

   // Configure ADC A1 pin
   SYSCFG2 |= ADCPCTL8;

   // Configure XT1 oscillator
   P4SEL0 |= BIT1 | BIT2;                           // P4.2~P4.1: crystal pins

   // Disable the GPIO power-on default high-impedance mode to activate
   // previously configured port settings
   PM5CTL0 &= ~LOCKLPM5;

   FRCTL0 = FRCTLPW | NWAITS0;                        // Set number of FRAM waitstates to 0

   CSCTL4 = SELMS__DCOCLKDIV | SELA__XT1CLK;             // Set ACLK = XT1; MCLK = SMCLK
= DCO

   do
   {
      CSCTL7 &= ~(XT1OFFG | DCOFFG);                  // Clear XT1 and DCO fault flag
      SFRIFG1 &= ~OFIFG;
   }while (SFRIFG1 & OFIFG);                         // Test oscillator fault flag

   //CONFIGURE BUTTON
   P1DIR &= ~BIT2; //input P1.2
   P1REN |= BIT2; //enable pull-up resistor for P1.2
   P1OUT |= BIT2; //enable pull-up resistor for P1.2

   P1IES &= ~BIT2; //interrupt on falling edge "&="
   P1IE  |= BIT2; //P1 interrupt enable for bit2
   P1IFG &= ~BIT2; //P1 interrupt flag clear


   // Configure ADC - Pulse sample mode; ADCSC trigger
   ADCCTL0 |= ADCSHT_8 | ADCON + ADCMSC;                    // ADC ON,temperature
sample period>30us
   ADCCTL1 |= ADCSHP + ADCCONSEQ_2;                         // s/w trig, single ch/conv,
```

```
MODOSC
   ADCCTL2 |= ADCRES;                                  // 10-bit conversion results
   ADCMCTL0 |= ADCSREF_0 | ADCINCH_8;                  // ADC input ch A12 => temp sense
   ADCIE |=ADCIE0;                                     // Enable the Interrupt request for a completed
ADC_B conversion

   // Configure reference
   PMMCTL0_H = PMMPW_H;                                // Unlock the PMM registers
   PMMCTL0_L |= PMMREGOFF_L;                           // and set PMMREGOFF
   PMMCTL2 |= INTREFEN;                                // Enable internal reference and temperature
sensor
   __delay_cycles(400);                                // Delay for reference settling

// LCD

   // Configure LCD pins
   SYSCFG2 |= LCDPCTL;                                 // R13/R23/R33/LCDCAP0/LCDCAP1 pins
selected

   LCDPCTL0 = 0xFFFF;
   LCDPCTL1 = 0x07FF;
   LCDPCTL2 = 0x00F0;                                  // L0~L26 & L36~L39 pins selected

   LCDCTL0 = LCDSSEL_0 | LCDDIV_7;                     // flcd ref freq is xtclk

   //LCD Operation - Mode 2
   LCDVCTL = LCDCPEN |                                 //Enable charge pump
         LCDSELVDD |                                   //Internally connect to Vcc
         VLCD_8 |                                      //internal 3.08v
        (LCDCPFSEL0 | LCDCPFSEL1 | LCDCPFSEL2 | LCDCPFSEL3);//Charge pump frequency
selection 256Hz

   LCDMEMCTL |= LCDCLRM | LCDCLRBM;                    // Clear LCD memory

   LCDCSSEL0 = 0x000F;                                 // Configure COMs and SEGs
   LCDCSSEL1 = 0x0000;                                 // L0, L1, L2, L3: COM pins
   LCDCSSEL2 = 0x0000;



   LCDBLKCTL = LCDBLKPRE__512 |                        //Divide xtclk by 512
         LCDBLKMOD_3;                                  //Switch between memory contents of LCDM and
LCDB
```

```
 LCDCTL0 |= LCD4MUX | LCDON;


ADCCTL0 |= ADCENC;
_EINT();
//LPM0;                    // Turn on LCD, 4-mux selected

 while(1){


 LCDM0 = 0x21;                          // L0 = COM0, L1 = COM1
 LCDM1 = 0x84;                          // L2 = COM2, L3 = COM3

 showChar('P', pos1);
 showChar('R', pos2);
 showChar('E', pos3);
 showChar('S', pos4);
 showChar('S', pos5);

 LCDBM0 = 0x21;
 LCDBM1 = 0x84;

 showCharB('B', pos1);
 showCharB('U', pos2);
 showCharB('T', pos3);
 showCharB('T', pos4);
 showCharB('O', pos5);
 showCharB('N', pos6);

 LPM0;
 buttonFlag = 1;

 LCDMEMCTL |= LCDCLRM | LCDCLRBM;
                  // Clear LCD memory
 LCDM0 = 0x21;                          // L0 = COM0, L1 = COM1
 LCDM1 = 0x84;
 LCDBM0 = 0x21;
 LCDBM1 = 0x84;


 dataVal = 100 - ((mVal*100)/1024);
 displayData(dataVal);
```

```
   LPM0;
   }
}

// ADC interrupt service routine
void ADC_ISR(void) __interrupt[ADC_VECTOR]{
 if(i < 400){
 ADC_Result[i] = ADCMEM0;
 i++;
 __delay_cycles(10000);
 }
 else{//input code here for when we're done taking our samples
 ADCCTL0 &= ~ ADCENC;

   for(i = 0; i < 400; i++)
   {
    if(ADC_Result[i] > mVal){
      mVal = ADC_Result[i];
    }
   }

 i=0;

 LPM0_EXIT;

 }
 P1IV = 0;
 //while(1);
}

void Button(void) __interrupt[PORT1_VECTOR]{ //start sampling when button pressed
//Check if button press happened
  if (P1IV == P1IV_P1IFG2) //Button press interrupt present for P1.2
  {
   if(buttonFlag == 0){
    ADCCTL0 |= ADCSC + ADCENC; //start conversion

    LCDMEMCTL |= LCDCLRM | LCDCLRBM;
    LCDM0 = 0x21;                              // L0 = COM0, L1 = COM1
    LCDM1 = 0x84;
    showChar('M', pos1);
    showChar('E', pos2);
```

```
    showChar('A', pos3);
    showChar('S', pos4);

    LCDBM0 = 0x21;
    LCDBM1 = 0x84;
    showCharB('M', pos3);
    showCharB('E', pos4);
    showCharB('A', pos5);
    showCharB('S', pos6);
   }
   else{
  buttonFlag = 0;
   }

  }
  LPM0_EXIT;
  P1IV = 0;


  __no_operation();
}

/* PROGRAM NOTES
program uses multiple sample single channel activated via button to take x samples then stop.

samples for about 6.5 seconds right now.

current issues:
phantom break point issue when placing break points sometimes?

*/
```